Task 2 Project Report

Project overview

This project implements and verifies reusable digital logic in synthesizable Verilog.
The work covers sequential circuits, counters, parameterized RTL, finite state machines, clock and reset behavior, simulation, and a timed traffic-light controller.

Objectives

- Design sequential logic with synchronous and asynchronous reset options.
- Implement common counters and parameterized reusable RTL blocks.
- Build Moore and Mealy controllers and practice state-based design.
- Verify behavior with self-checking testbenches and inspectable waveform traces.

Development environment

Ubuntu 26.04.1 LTS under WSL; Icarus Verilog 12.0; GTKWave 3.3.126; Git 2.53.0.
The project is maintained in a public GitHub repository on its main branch.

Repository

https://github.com/vishnuvamsi396-gif/VLSI-Design-Task-2

DESIGN AND IMPLEMENTATION

Sequential circuits

The sequential RTL includes a D flip-flop, an 8-bit register, and a shift register. Synchronous reset is sampled on the active clock edge. Asynchronous reset acts independently of that edge. Edge-triggered state updates use nonblocking assignments.

Counters and parameterized blocks

Implemented counters: 4-bit up and down, Mod-10, 4-bit ring, and 4-bit Johnson. Parameterized register, counter, and shift-register modules support multiple widths. The self-checking parameter testbench exercises both 4-bit and 8-bit instances.

Finite state machines

Traffic-light examples include a Moore controller and a Mealy controller with an emergency input. The vending machine accepts 5-unit and 10-unit coins for a 15-unit product. The elevator controller handles requests among three floors.
The mini-project is an autonomous Moore traffic-light controller with configurable RED, GREEN, and YELLOW durations. Its outputs depend on the registered state.

Design documentation

SVG diagrams show the RTL overview, traffic-light FSM, vending FSM, elevator FSM, and a conceptual clock/reset timing illustration. Counter sequences and reset behavior are described in docs/design_notes.md.

SIMULATION AND SUBMISSION STATUS

Verification results

All seven self-checking testbenches passed with Icarus Verilog 12.0 in Ubuntu/WSL.

- Sequential logic and synchronous/asynchronous reset behavior.
- Up, down, Mod-10, ring, and Johnson counter sequences.
- Parameterized register, counter, and shift-register operation at 4 and 8 bits.
- Moore and Mealy traffic-light behavior and timed traffic-light mini-project.
- Vending-machine and elevator state transitions.
- Reset timing around rising clock edges.

The corresponding seven simulator logs are in simulation/results/ and seven VCD traces are in simulation/vcd/. Run all tests with: bash scripts/run_all.sh

Remaining evidence

GTKWave screenshots still need to be captured after visually inspecting the VCD traces. The screenshots are not represented as complete in this report or checklist until they are added to waveforms/. The submission checklist is docs/requirements_checklist.md.

Conclusion

The RTL and verification flow are organized for reuse. The passing test suite provides simulation evidence for the implemented circuits and controllers.